

A shared e-commerce platform for student-run micro-businesses at MFU, with an AI advisor that turns the platform's own data into recommendations for sellers.

1 · Research

Does recommendation framing change the 7-day action rate?

2 · Coursework

Auth, CRUD, dashboard, cloud, external API

3 · Product

Real sellers, one semester

26

FEATURES

36

BACKLOG ITEMS

47

NFRS

19

TEST CASES

2

ADRS ACCEPTED

3

DECISIONS OPEN

FEATURE LIST — 8 GROUPS

A	Accounts & access	auth
B	Store & catalogue — products, photos , price, stock	CRUD
C	Buyer & orders	
D	Seller dashboard	dashboard
E	AI advisor — grounded recommendations	
F	Telemetry & experiment infrastructure	
G	Analysis	
H	Platform — deployment, external API	cloud API open

Every user action writes an event. No event, no data — and none of it can be backfilled.

DOCUMENT SET

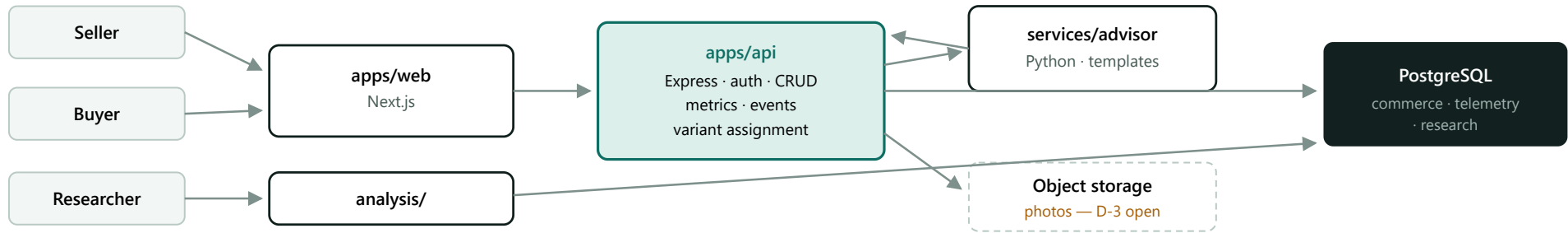
- 01 requirements · non-functional-requirements · product-backlog · feature-list
- 02 architecture · database-schema · api-spec · detailed-design
- 02 UJ-01 journey · PR-01 wireframe + prototype · ADR-0001 · ADR-0002
- 03 hypotheses · analysis-plan (thesis)
- 04 TS-01 test spec

TRACEABILITY

REQ-B3 → PB-10 → FEAT-B4 → [product.photo_added](#) → TS-01-07

One chain per feature. The same event is the advisor's action_type target and step 9 of the journey.

EVERY USER ACTION PASSES THROUGH ONE API



THREE PLACEMENT RULES

- **Events** are written by the API, inside the action's transaction. Never by the browser.
- **Assignment** happens on the server, once, persisted. Unique per unit.
- **Metrics** live in one module, read by both the dashboard and the advisor.

STACK

web	Next.js, TypeScript
api	Express, TypeScript
db	PostgreSQL via Prisma
advisor	Python — templates, no LLM (ADR-0002)
analysis	Python
deploy	Cloud, managed Postgres — provider open (D-1)
action + event → one transaction → commit	

DATA MODEL — THREE ZONES

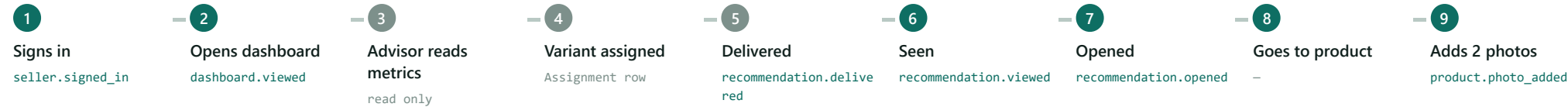
ZONE	TABLES	POLICY
Commerce	User · Store · Product · ProductPhoto · Order · OrderItem	reshape freely
Telemetry	Event	append-only
Research	Experiment · Variant · Assignment · Recommendation	ask first

CONVENTIONS

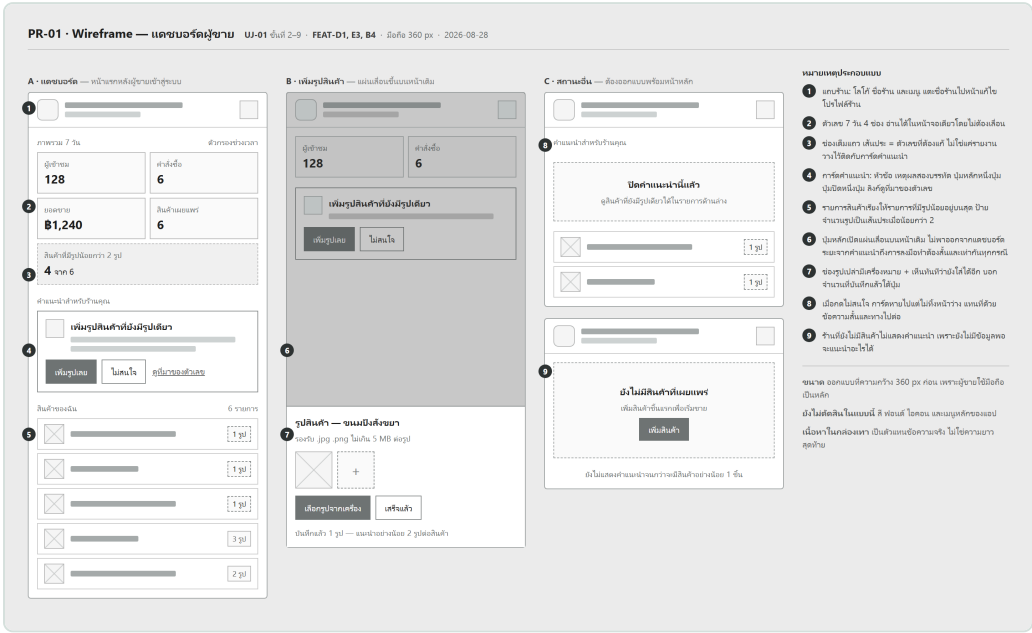
- **Event has no foreign keys** — a deleted product must not erase the measurement
- **Money in integer satang** — float totals stop adding up
- **is_seed on every table** — demo rows excluded everywhere
- **OrderItem stores price and name** — later edits never rewrite history

acted = event(store, action_type) <= delivered_at + 7 days

THE JOURNEY THE THESIS MEASURES — AND THE EVENT AT EACH STEP



WIREFRAME — PR-01, SELLER DASHBOARD



ENDPOINTS ON THIS PATH

GET	/dashboard/metrics	dashboard.viewed
GET	/recommendations/current	recommendation.delivered
POST	/recommendations/:id/dismiss	recommendation.dismissed
POST	/products/:id/photos	product.photo_added

MEASUREMENT

delivered_at + action_type + store_id
→ event within 7 days → acted 0/1
→ rate per variant

TESTS & DECISIONS

TS-01	Assignment stable · snapshot frozen · 7-day boundary exact · no personal data · events immutable
ADR-0001	Append-only events, written in the action's transaction
ADR-0002	Advisor copy from templates, not an LLM
Open	D-1 hosting · D-2 auth · D-3 object storage